

GALNET Documentation

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1 Introduction

The study of galaxy structural parameters is fundamental to understanding the formation and evolution of galaxies across cosmic time. Parameters such as magnitude, effective radius, Sérsic index, and ellipticity provide direct insight into the galaxy morphology. Traditionally, these quantities have been derived using computationally intensive model fitting techniques, which can be slow and sensitive to noise, background variations, and point-spread function (PSF) effects. To address these challenges, we present GALNET, a convolutional neural network (CNN) designed to estimate galaxy structural parameters efficiently and robustly. The tool has been developed as part of the LSST in-kind contribution (ITA-INA-S9). GALNET is trained using simulated datasets generated from real astronomical backgrounds and PSFs, enabling the network to learn realistic observational conditions. Once trained, GALNET can predict galaxy parameters directly from images, bypassing the need for iterative fitting procedures.

This documentation provides an overview of the GALNET workflow, including dataset generation, model training, prediction, and visualization. We describe the structure of the input data, the functions implemented in the codebase, and the meaning of each column in the generated CSV files. The goal is to offer a reproducible and transparent framework for researchers who

wish to apply GALNET to their own datasets or extend its functionality for future astronomical surveys.

2 Setting up an Environment for GALNET

To set up the GALNET environment, follow the steps below:

2.1 Clone the Repository

First, obtain the source code from GitHub:

```
git clone https://github.com/aiswarya-unni/Galaxy-Structural-Parameters.git
cd Galaxy-Structural-Parameters/galnet
```

2.2 Create the Conda Environment

Create a new Conda environment named `galnet` with Python 3.10:

```
conda create -n galnet python=3.10
conda activate galnet
```

2.3 Install Dependencies

Install all required Python packages listed in `requirements.txt`:

```
pip install -r requirements.txt
```

2.4 Verify Installation

You can confirm that the environment is active by listing all Conda environments:

```
conda env list
```

The `galnet` environment should appear marked as active.

2.5 Download the Dataset

The dataset required to run GALNET is hosted on Google Drive. It can be downloaded using the `gdown` utility:

```
pip install gdown
gdown "https://drive.google.com/uc?id=150aGl6HFyipI5PyuIno9xNSY31p1noSA"
```

After downloading, extract the archive and place the contents into:

```
./galnet/data/
```

3 Repository Structure

The GALNET repository is organized as follows:

```
Galaxy-Structural-Parameters/  
galnet/  
  data/                                # Downloaded separately  
    real/  
      image/  
      psf/  
      background.csv  
    train/  
    validation/  
    test/  
    train_data.csv  
    validation_data.csv  
    test_data.csv  
  GaLNet/  
    cmodel.py  
    train.py  
    plot_test.py  
    pred_test.py  
    fig/  
    result/  
  model/  
    weights.h5  
  galaxy.py  
  requirements.txt  
  .gitignore
```

Description of Key Directories

- `data/` — Contains real backgrounds, PSFs, and generated datasets (train, validation, test).
- `GaLNet/` — Core CNN implementation, training, prediction, and plotting scripts.

- `model/` — Stores trained model weights.
- `galaxy.py` — Script to generate simulated galaxy datasets.
- `requirements.txt` — Python dependencies.

4 Data Generation

Once the dataset has been downloaded from Google Drive and correctly placed in the `./galnet/data/` directory, the data generation process can begin.

4.1 Input Data

The `image/` and `psf/` folders inside `data/real/` contain FITS files from LSST DP1. These represent real background images and their corresponding point-spread functions (PSFs). The file `background.csv` links each background image with its PSF and sky coordinates. An example entry is:

```
image_name,psf_name,ra,dec
image_i_0.fits,psf_i_0.fits,52.912555625468954,-27.977151877119923
```

The file `background.csv` contains metadata linking each background image to its corresponding PSF and sky coordinates. The columns are defined as follows:

- `image_name`: The background FITS file containing the real sky image into which simulated galaxies are injected.
- `psf_name`: The PSF FITS file associated with the background image, describing how a point source is blurred by the telescope optics and atmosphere.
- `ra`: The Right Ascension of the background image, expressed in degrees. This is the sky coordinate analogous to longitude.
- `dec`: The Declination of the background image, expressed in degrees. This is the sky coordinate analogous to latitude.

4.2 Running the Generator

To generate datasets, run:

```
python galaxy.py
```

The dataset mode is controlled by the `mode` parameter inside `galaxy.py`:

```
mode = "train"      # Training dataset
mode = ""           # Validation dataset
mode = "test"       # Test dataset
```

Depending on the chosen mode, the script will create the corresponding directory (`train/`, `validation/`, or `test/`) inside `galnet/data/`, along with the associated CSV file (`train_data.csv`, `validation_data.csv`, or `test_data.csv`).

4.3 Parameter Ranges

The simulated galaxies are generated with parameters drawn from the following ranges:

Parameter	Range	Units	Distribution
x_{cen}	−0.4–0.4	arcsec	uniform
y_{cen}	−0.4–0.4	arcsec	uniform
mag	16–25	...	uniform
R_{e}	0.2–4	arcsec	normal ($\log R_{\text{e}}$) ($\mu = 0.7$, $\sigma = 1$)
q	0.2–1.0	...	normal ($\mu = 0.8$, $\sigma = 0.3$)
PA	0.0–180.0	deg	uniform
n	0.1–8.0	...	$F(n = 30, d = 5)$

Table 1: Parameters for Simulating the Training Sample

4.4 Configurable Parameters

Users can adjust several parameters in `galaxy.py` to suit their needs:

- **Number of galaxies:** Controlled by the variables `train_num`, `val_num`, and `test_num`, which set the number of samples for the training, validation, and test datasets respectively.
- **Signal-to-noise ratio (SNR):** By default, galaxies with SNR greater than 35 are used.
- **Pixel scale:** The LSST pixel size is set to 0.2 arcsec/pixel.
- **Image and PSF dimensions:** Controlled by `pix_num` and `psf_pix_num`.
- **Zero-point magnitude:** Defined by the variable `mag_zero`, which sets the photometric calibration constant used to convert flux values into magnitudes. For LSST DP1, this is typically set to 31.4.

4.5 Sérsic Galaxy Model

The intrinsic surface brightness distribution $I(x, y)$ of each simulated galaxy is described using a Sérsic profile. The Sérsic surface brightness distribution is defined as

$$I(x, y) = I_e \exp \left\{ -b_n \left[\left(\frac{R(x, y)}{R_e} \right)^{\frac{1}{n}} - 1 \right] \right\} \quad (1)$$

where I_e is the surface brightness at the effective radius R_e , n is the Sérsic index, and b_n is the Sérsic constant associated with the Sérsic index. The elliptical radius $R(x, y)$ is given by

$$R(x, y) = \sqrt{\frac{x'^2}{q} + q, y'^2}, \quad (2)$$

where q is the minor-to-major axis ratio of the galaxy. The coordinates (x', y') are the rotated coordinates obtained by translating the galaxy coordinates to the galaxy centre $(x_{\text{cen}}, y_{\text{cen}})$ and rotating them by the position angle PA. They are defined as

$$\begin{aligned} x' &= (x - x_{\text{cen}}) \cos \text{PA} + (y - y_{\text{cen}}) \sin \text{PA} \\ y' &= -(x - x_{\text{cen}}) \sin \text{PA} + (y - y_{\text{cen}}) \cos \text{PA} \end{aligned} \quad (3)$$

Here, $(x_{\text{cen}}, y_{\text{cen}})$ denotes the centre of the galaxy, R_e is the effective (half-light) radius, I_e is the surface brightness at R_e , PA is the position angle, q is the axis ratio, and n is the Sérsic index. In the simulation, the position angle is defined with respect to the sky coordinate frame, with the positive direction taken from north to east.

The Sérsic profile is evaluated on the image grid using these parameters to generate the intrinsic galaxy light distribution. The resulting galaxy model is subsequently convolved with the corresponding point-spread function (PSF) and combined with a real astronomical background image to produce the final simulated galaxy image used by GALNET.

4.6 Final Simulated Galaxy Output

The simulated galaxy is stored as a multi-extension FITS file. Each Header Data Unit (HDU) contains a different component of the simulation: The HDU 0 contains the simulated Sérsic profile of the galaxy, convolved with the PSF, the HDU 1 has the real background image cutout, representing sky noise and structure, finally the HDU 2 the point-spread function (PSF) used in the simulation.

Similarly, the corresponding output CSV with simulated parameters are defined as follows:

- **name:** The filename of the simulated galaxy image.
- **mag:** Magnitude of the galaxy, representing its brightness.
- **xcen:** X-coordinate of the galaxy center relative to the image, in arcseconds.
- **ycen:** Y-coordinate of the galaxy center relative to the image, in arcseconds.
- **R_eff:** Effective radius of the galaxy, i.e., the radius enclosing half of the total light, in arcseconds.
- **pa:** Position angle of the galaxy, measured in degrees from north through east.
- **q:** Axis ratio of the galaxy (b/a), describing its ellipticity.
- **n:** Sérsic index, controlling the concentration of the galaxy light profile.
- **SNR:** Signal-to-noise ratio of the galaxy generated.

An example row is shown below:

```
name,mag,xcen,ycen,R_eff,pa,q,n,SNR  
0.fits,16.3955,-0.0509,0.2815,2.8829,39.672,0.4882,4.7027,8816.3828
```

5 Training

The GALNET training procedure uses the simulated galaxy datasets generated by `galaxy.py`. The training and validation samples are stored in `data/train/` and `data/validation/`, respectively, with their corresponding parameter catalogues stored in `train_data.csv` and `validation_data.csv`. The neural network is trained to predict the structural parameters of a galaxy directly from its simulated image and the corresponding point-spread function (PSF). After the dataset has been generated and the GALNET environment has been activated, the training can be started from the `galnet/` directory using:

```
python GaLNet/train.py
```

5.1 Training Configuration

The main training parameters are defined in `GaLNet/train.py`. The default configuration used for GALNET is:

Parameter	Value
Number of training galaxies	100,000
Number of validation galaxies	20,000
Number of epochs	150
Batch size	128
Learning rate	10^{-3}
Image size	81×81 pixels
PSF size	35×35 pixels
Input size	$116 \times 81 \times 1$
Optimizer	Adam
Loss function	Huber loss
Metrics	MSE, MAE, R^2

Table 2: Default training configuration used in GALNET v1.

The number of training and validation samples can be modified in `train.py` if a different dataset size is required.

5.2 Input Data Preparation

Each simulated galaxy is stored as a multi-extension FITS file. For training, the galaxy image and PSF are read separately from the FITS file. The galaxy image is obtained from HDU 0, while the PSF is obtained from HDU 2. The GALNET architecture is given below:

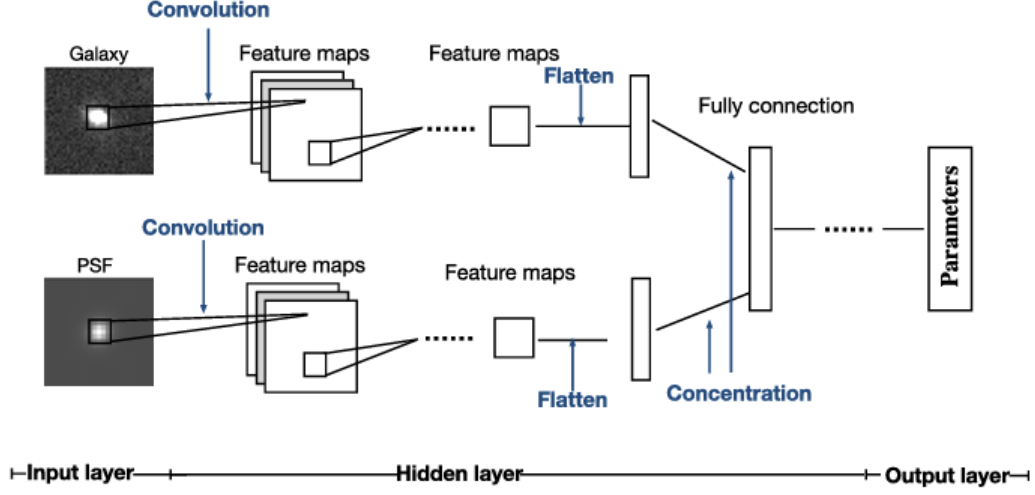


Figure 1: The architectural pipeline of GALNET, showing the parallel processing of the galaxy cutout and the PSF frame

The image and PSF are scaled before being supplied to the neural network. The PSF has a smaller spatial extent than the galaxy image. Therefore, the PSF is padded to match the spatial dimensions required for concatenation. An additional channel dimension is introduced before the data are passed to the CNN. The target galaxy parameters are also normalized before training. This allows the different structural parameters, which have different physical units and numerical ranges, to be represented on comparable scales.

The following parameter ranges are used for normalization:

Parameter	Minimum	Maximum
Magnitude	16	25
x_{cen}	-0.4	0.4
y_{cen}	-0.4	0.4
R_{e}	0.2	4.0
n	0.1	8.0
e_1	-0.67	0.67
e_2	-0.67	0.67

Table 3: Parameter ranges used for normalization during training.

The ellipticity components e_1 and e_2 are calculated from the axis ratio and position angle of the simulated galaxy as below.

$$e_1 = \frac{1-q}{1+q} \cos\left(2\frac{\text{PA} \cdot \pi}{180}\right), \quad e_2 = \frac{1-q}{1+q} \sin\left(2\frac{\text{PA} \cdot \pi}{180}\right) \quad (4)$$

These normalized parameters are used as the target values during network training.

5.3 Training the CNN

The CNN architecture used in GALNET is implemented in `GalNet/cmodel.py` and is initialized using the input shape defined in `train.py`:

```
inpt = (116, 81, 1)
model = cmodel.CNN(inpt)
```

The model is compiled using the Adam optimizer with a learning rate of 10^{-3} . The Huber loss function is used as the training objective. Mean squared error (MSE), mean absolute error (MAE), and the coefficient of determination (R^2) are monitored during training and validation.

The model is trained using the following configuration:

```
epochs = 150
batch_size = 128
learning_rate = 1e-3

model.compile(
```

```

    loss=huber_loss,
    optimizer=Adam(learning_rate=learning_rate),
    metrics=[mse, mae, r_square]
)

```

The validation dataset is supplied during training to monitor the performance of the network on galaxies that are not used to update the network weights. During training, the model weights are monitored using a Keras `ModelCheckpoint`. The trained weights are saved to:

```
./model/weights.h5
```

The saved weights can subsequently be loaded by `GaLNet/pred_test.py` for prediction on the independent test dataset. This separates the computationally expensive training stage from the prediction stage and allows the trained model to be reused without retraining.

6 Prediction

After the network has been trained, galaxy structural parameters can be estimated for an independent test sample using `GaLNet/pred_test.py`. The prediction script loads the trained model weights, reads the test dataset, performs inference with the CNN, and saves both the true and predicted parameters to a CSV file.

From the `galnet/` directory, execute:

```
python GaLNet/pred_test.py
```

The script loads the trained weights stored in

```
./model/weights.h5
```

and applies the network to all galaxies contained in the test dataset.

6.1 Input Data

The prediction stage requires:

- The trained model weights: `model/weights.h5`
- The simulated test images stored in `data/test/`
- The corresponding catalogue `data/test_data.csv`

6.2 Network Outputs

The CNN predicts seven normalized structural parameters: Magnitude (mag), X-coordinate of the galaxy centre (x_{cen}), Y-coordinate of the galaxy centre (y_{cen}), Effective radius (R_e), Ellipticity component (e_1), Ellipticity component (e_2) and Sérsic index (n). The predicted values are subsequently transformed back to their physical units using the inverse of the normalization applied during training. The prediction results are written to

`GaLNet/result/pred_test_para.csv`

This catalogue contains both the true simulated parameters and the GALNET predictions, allowing direct comparison of the model performance.

- **name:** The filename of the galaxy FITS image used for prediction.
- **mag_true:** The true magnitude of the simulated galaxy used as the reference value.
- **x_true:** The true X-coordinate of the galaxy centre relative to the image centre, in arcseconds.
- **y_true:** The true Y-coordinate of the galaxy centre relative to the image centre, in arcseconds.
- **re_true:** The true effective radius (R_e) of the galaxy, corresponding to the radius enclosing half of the total galaxy light, in arcseconds.
- **e1_true:** The true first ellipticity component of the galaxy, computed from the axis ratio and position angle.
- **e2_true:** The true second ellipticity component of the galaxy, computed from the axis ratio and position angle.
- **n_true:** The true Sérsic index of the galaxy, describing the concentration of the light profile.
- **mag_cnn:** The magnitude predicted by GALNET.
- **x_cnn:** The X-coordinate of the galaxy centre predicted by GALNET, in arcseconds.
- **y_cnn:** The Y-coordinate of the galaxy centre predicted by GALNET, in arcseconds.

- `re_cnn`: The effective radius (R_e) predicted by GALNET, in arcseconds.
- `e1_cnn`: The first ellipticity component predicted by GALNET.
- `e2_cnn`: The second ellipticity component predicted by GALNET.
- `n_cnn`: The Sérsic index predicted by GALNET.
- `SNR`: The signal-to-noise ratio of the simulated galaxy image used for prediction.

In addition to the prediction catalogue, the script automatically executes `plot_test.py`, which generates diagnostic comparison plots between the true and predicted parameters. These figures are stored in

`GaLNet/fig/`

and typically include one-to-one comparisons for magnitude, centroid position, effective radius, ellipticity components, and Sérsic index. These plots provide a visual assessment of the accuracy in the GALNET predictions.

Source Code and Documentation

The GALNET source code, installation instructions, datasets, and latest documentation are available at:

<https://github.com/aiswarya-unni/Galaxy-Structural-Parameters>